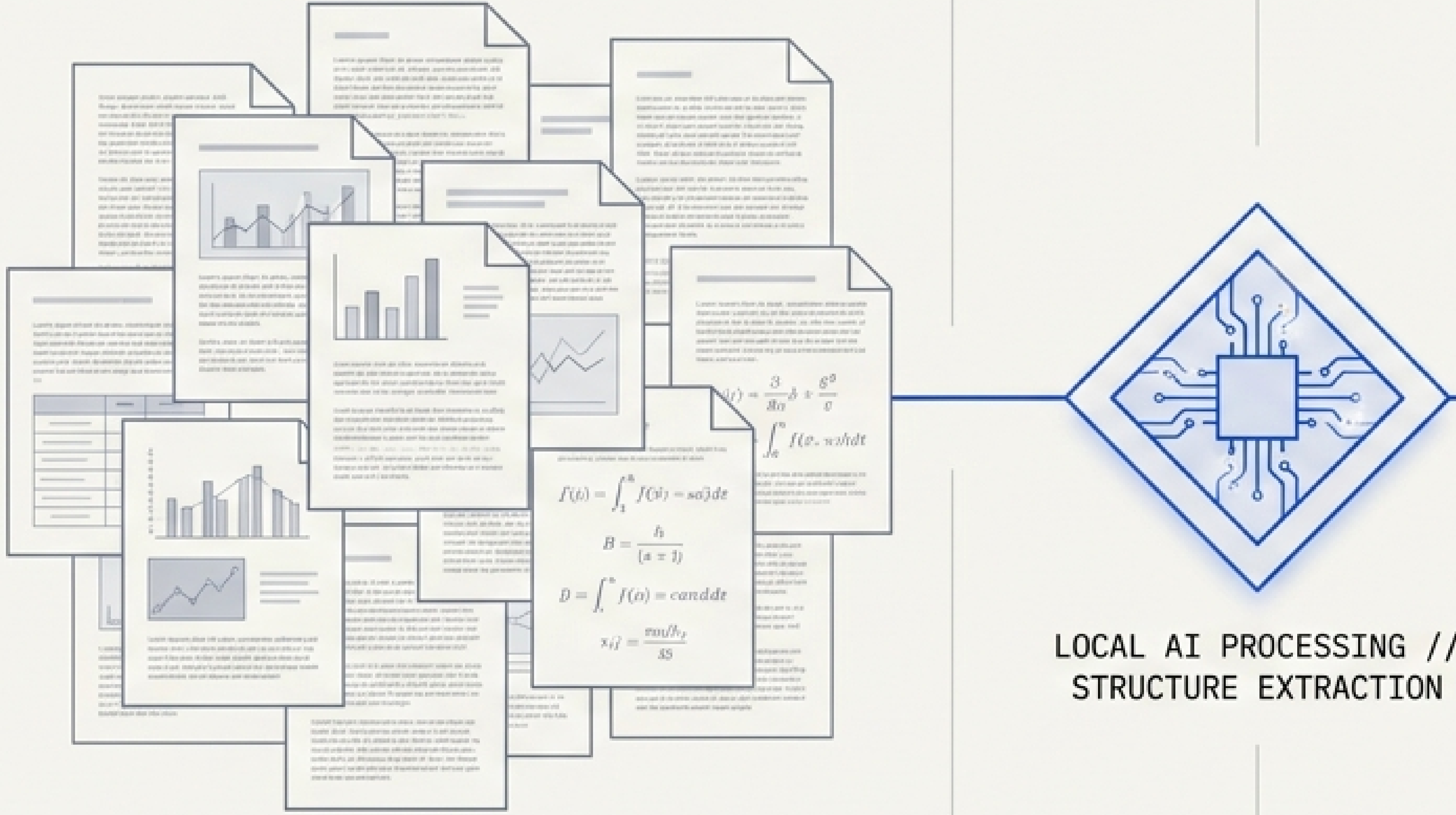
papers-to-table

A Local-First Pipeline for AI-Assisted Scientific Extraction

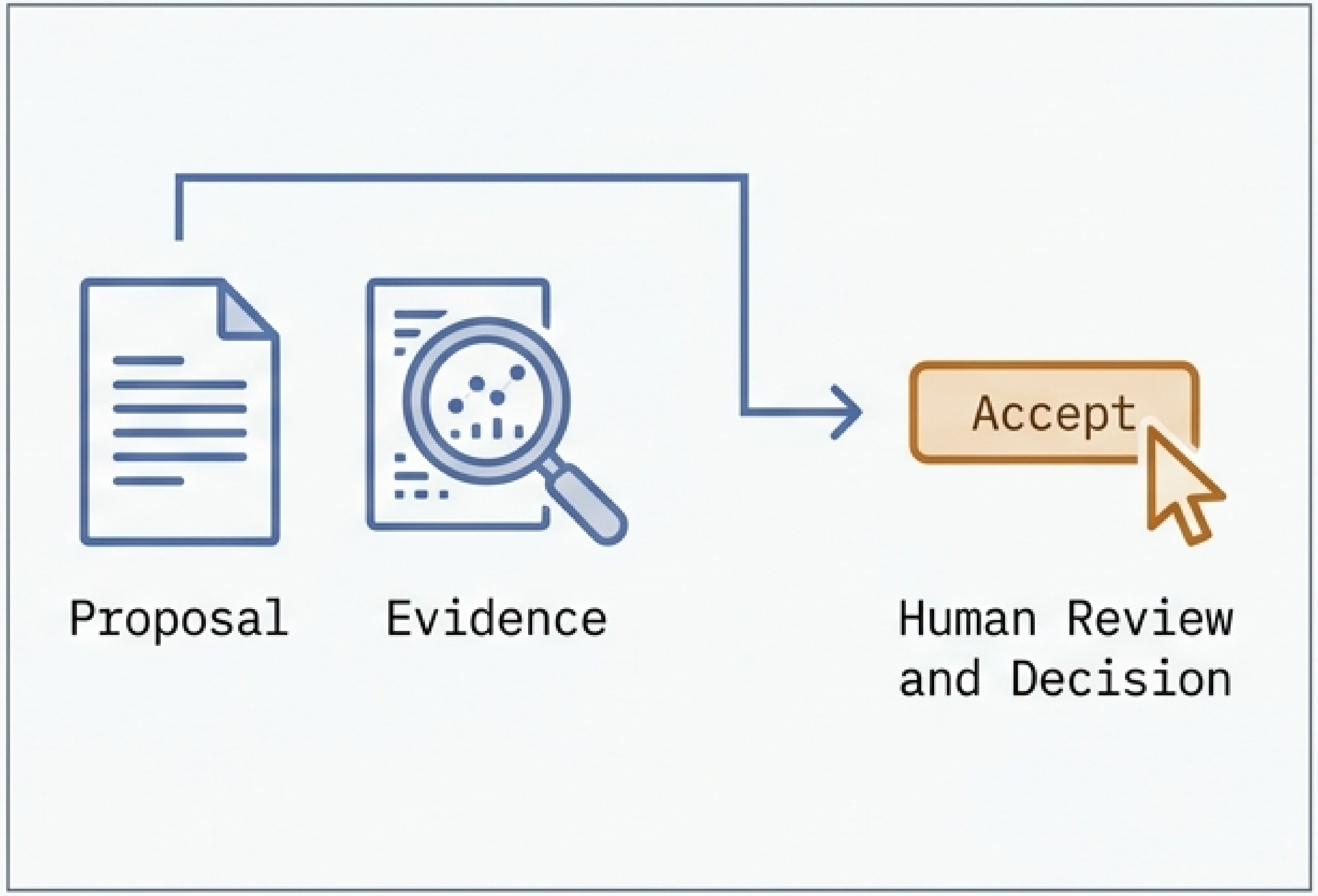


ID:	001		
PARAM_A:	45.2		
UNIT:	kg		
SOURCE_FIG:	28		
STATUS:	VERIFIED	HUMAN_REVIEW:	REQ
VALUE_B:	1.2e-5		
CONFIDENCE:	98.5%		
REVIEWER:	AMBER	HUMAN_REVIEW:	REQ
DATE:	2024-05-20		
NOTES:	VALIDATED		
VALUE_B:	1.2e-5		
CONFIDENCE:	98.5%	DECISION:	APPROVED
REVIEWER:	ANBER		
DATE:	2024-05-20		
NOTES:	VALIDATED	DECISION:	APPROVED

v. architecture-brief // overview & internals

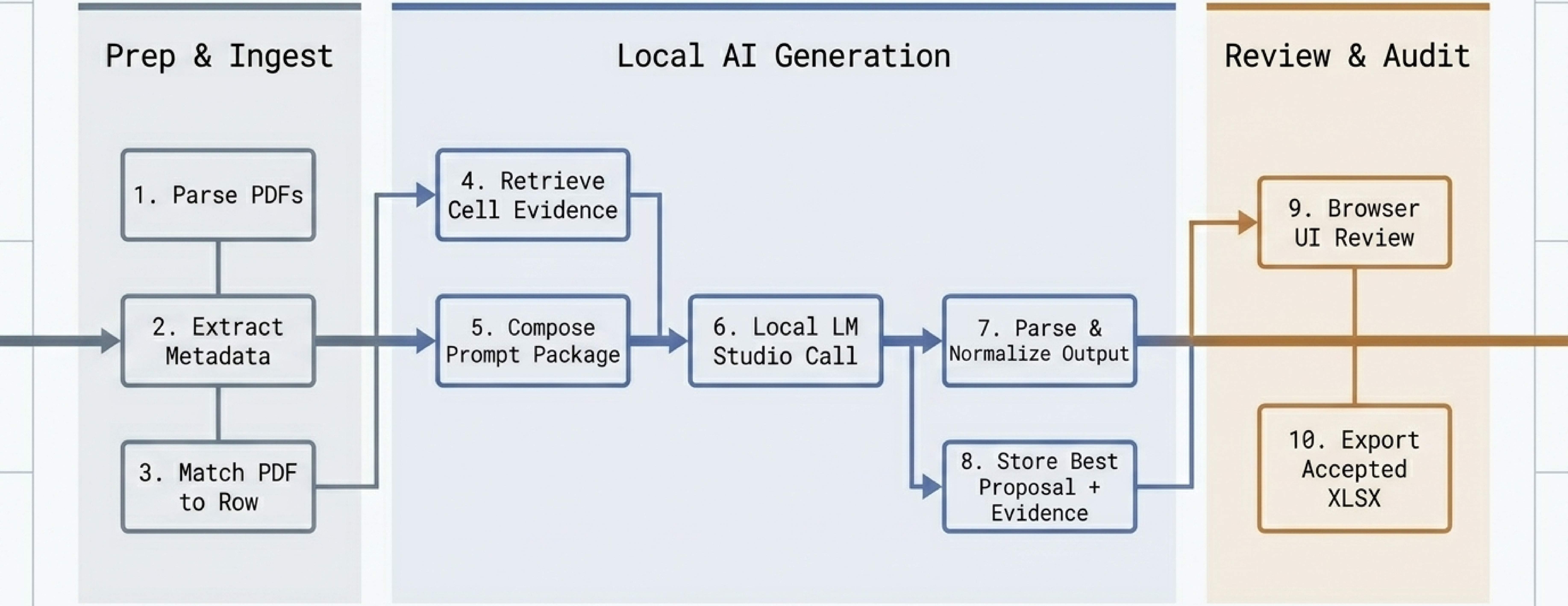
The Philosophy: Propose, Prove, Decide.

System Guardrail: The LLM never edits the spreadsheet directly.



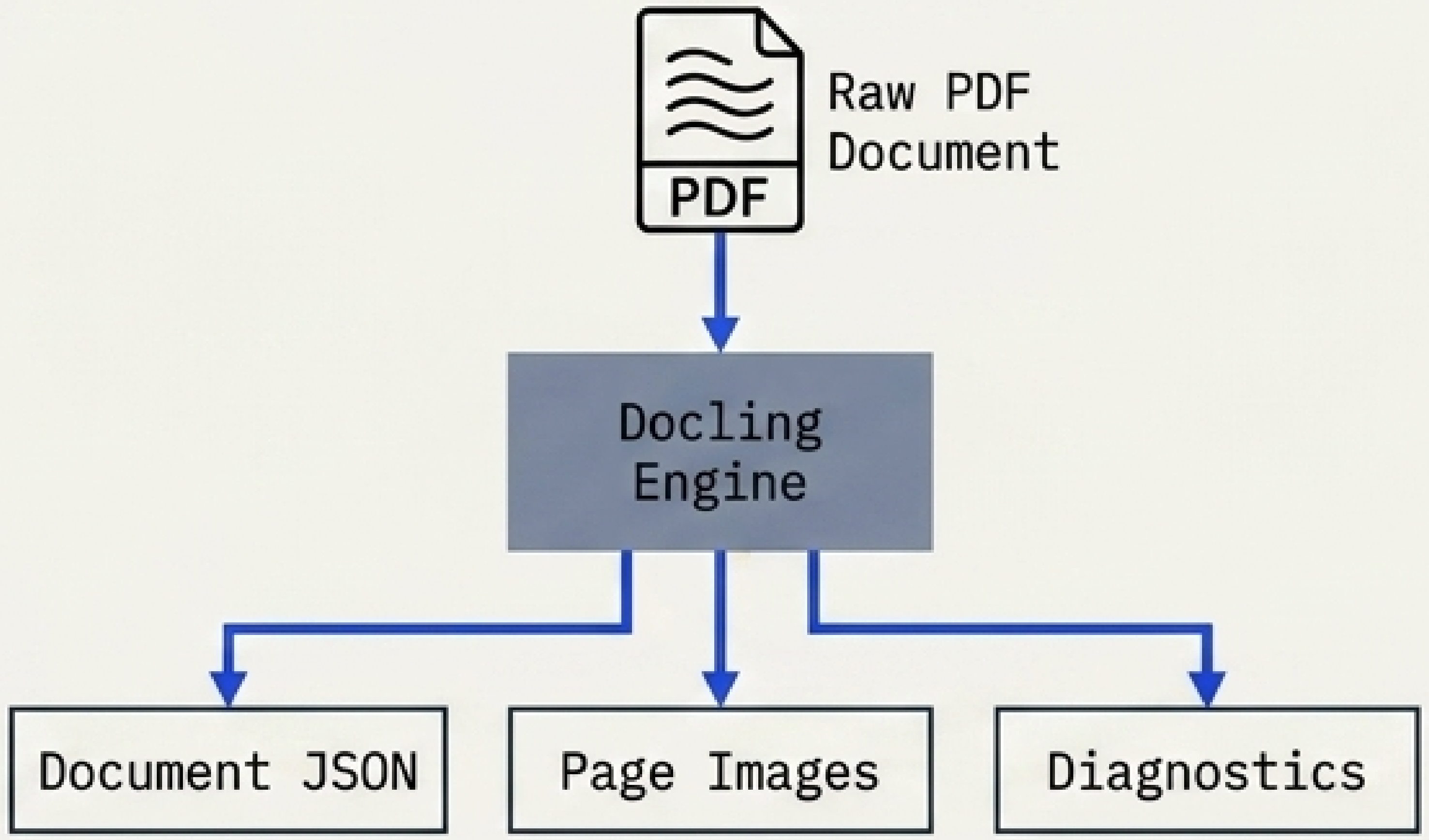
1. System proposes one best value per target cell.
2. System attaches auditable evidence.
3. Human reviewer decides what to accept.

The 10-Step Pipeline Architecture



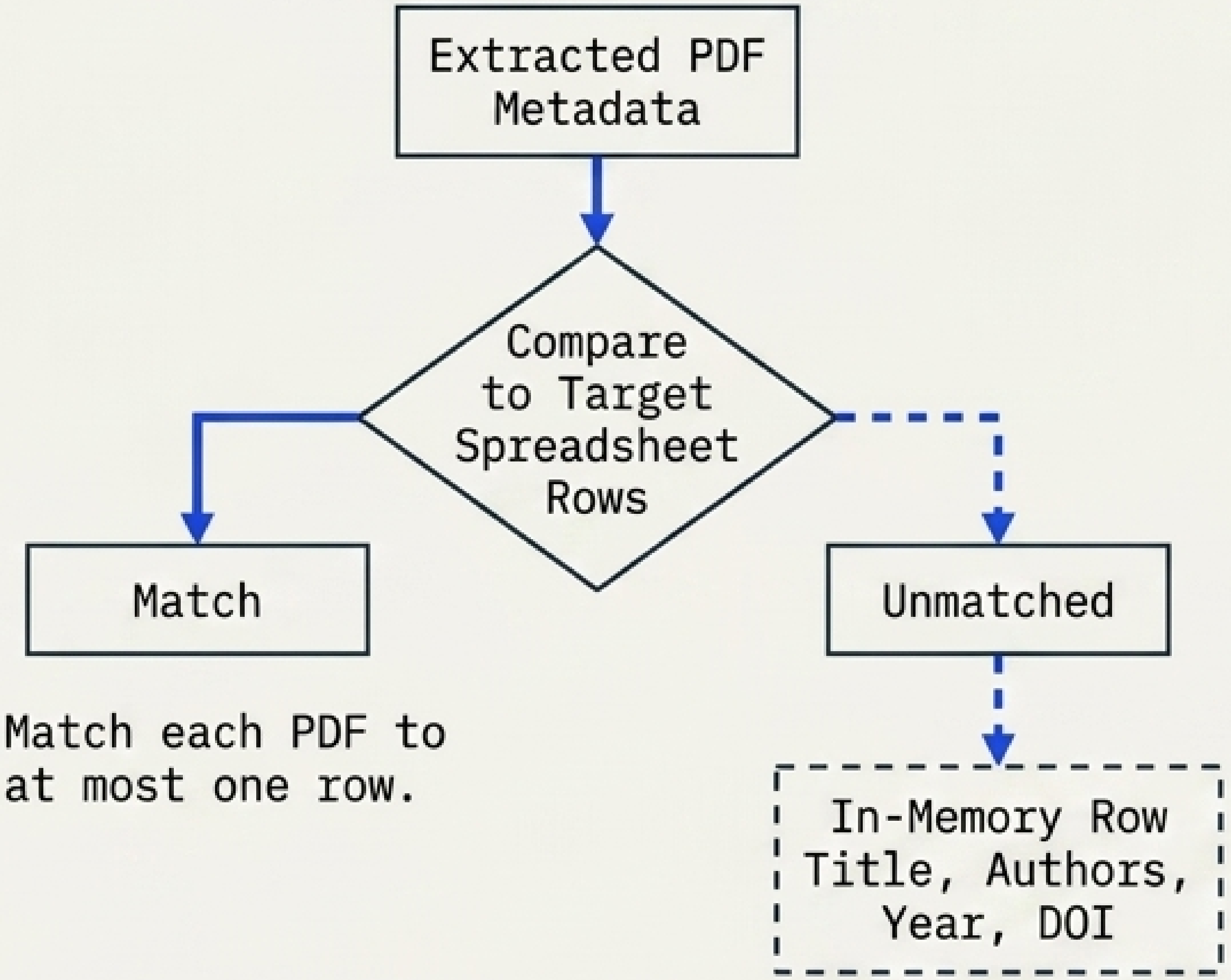
Establishing the Foundation: Parsing & Matching

Parsing (Backend: Docling)



Converts raw PDF into usable text, structure, and front matter. Fallbacks for OCR and degraded files.

Matching Strategy



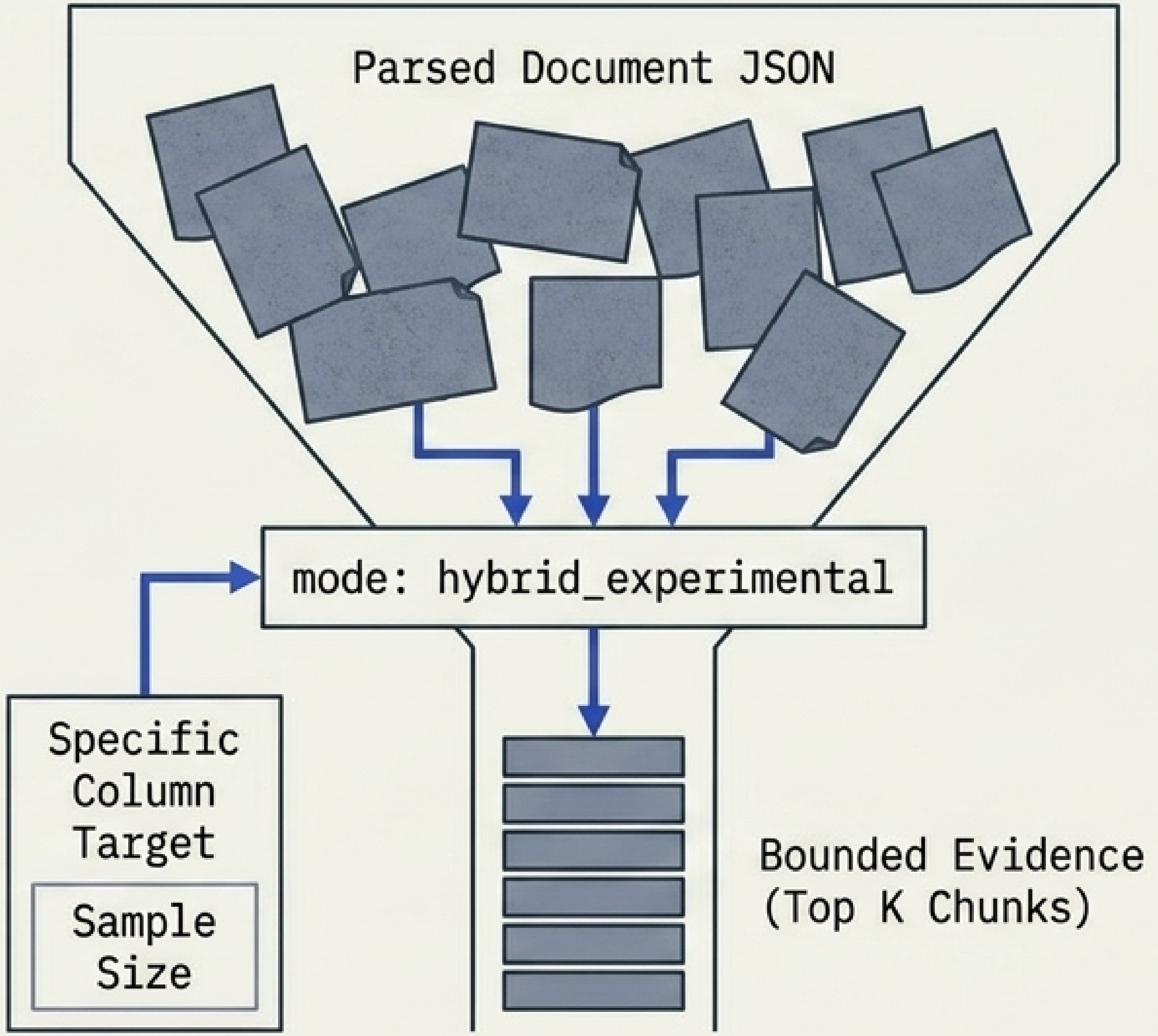
Ensures zero silent row overrides and explicit duplicate conflict resolution.

Precision Targeting: Evidence Retrieval

Row-aware and Column-aware.

Does not
not default to
whole-document
prompting.

Extracts Top K
chunks
(default: 12).



Eliminates context-window bloat, grounds the text model in specific paper geometry, and guarantees downstream human auditability.

The Assembly: Building the Prompt Package

Target Schema
Column name, description, field typing

+

Row Context
Explicit source paper identification

+

Retrieved Evidence
The Top K bounded chunks

+

Style Profile
Optional formatting guidance

=

Per-Cell Instruction Package

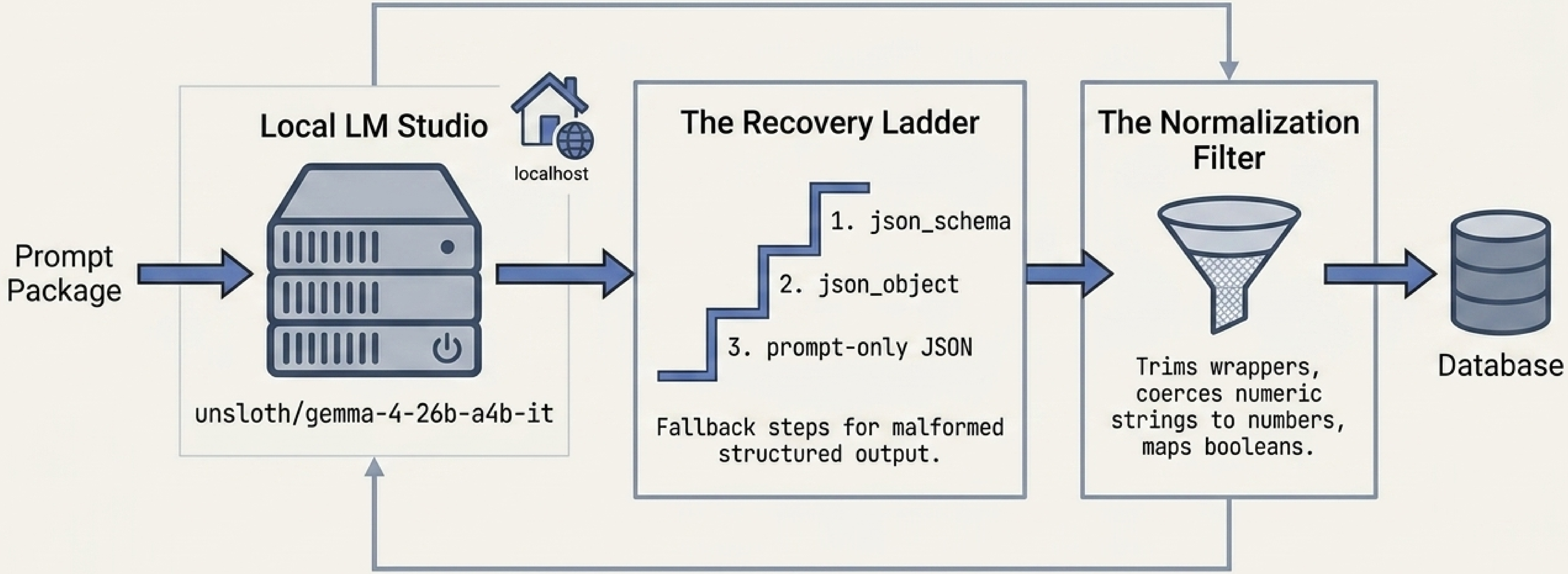
What is a Style Profile?

Inferred from existing filled cells via the text model to **match units**, **length**, and **format.** Guides the shape of the answer, NEVER provides the semantic content.

Format:
Unit: kg
Length: 2 words

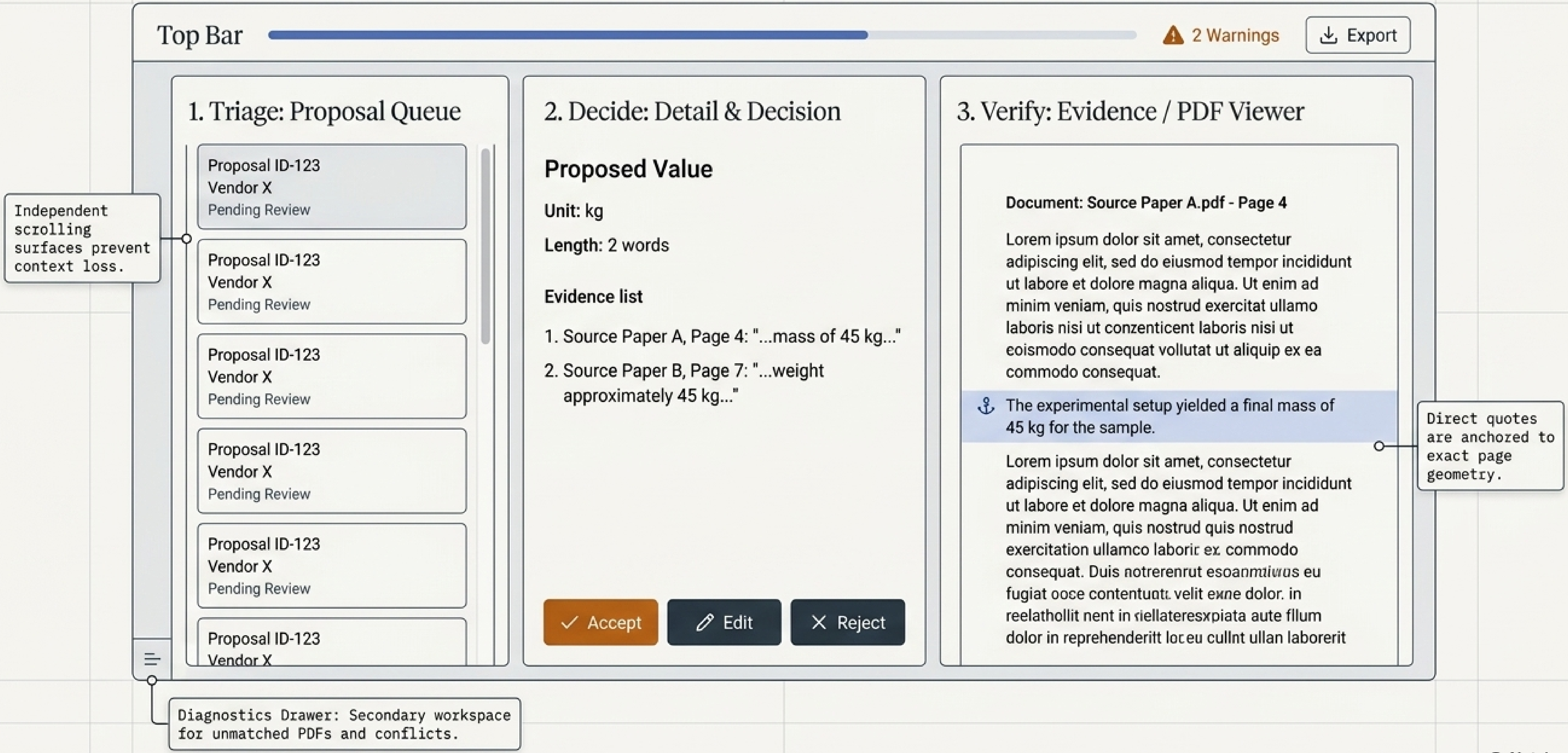
Semantic:
Value: 45

The Live Local Engine: LM Studio Integration



Zero cloud exposure. Outputs are deterministically normalized without semantic rewriting.

The 3-Panel Review Workspace



Worked Example: The Lifecycle of a Cell

Input Row: Smith et al. 2024 | **Target Column:** **Column:** Differentiation condition

Retrieval (Page 3) → Cells were differentiated by **LIF withdrawal** for 48 hours.

Prompting → Schema + Context + Evidence are sent to the local model.

LM Studio Proposal → **LIF withdrawal**, 48 h

Evidence Linking → UI attaches **Direct quote evidence** label and anchors geometry to Page 3.

Reviewer Action → Human inspects highlighted text and clicks **Accept**.

Result: **Target cell successfully populated**. Source workbook remains physically untouched.

The Final Output: Explicit & Auditable

The system ensures total provenance from raw PDF to the final cell value.